

Data & Things

(Spring 25)

Monday February 3

**Lecture 1: Introduction to the course,
data science, and Python for data science**

Jens Ulrik Hansen

Data & Things – spring 25

Harvard
Business
Review



DATA

Data Scientist: The Sexiest Job of the 21st Century

by Thomas H. Davenport and D.J. Patil

FROM THE OCTOBER 2012 ISSUE

<https://hbr.org/2012/10/data-scientist-the-sexiest-job-of-the-21st-century>



<https://www.economist.com/leaders/2017/05/06/the-worlds-most-valuable-resource-is-no-longer-oil-but-data>

Data & Things – spring 25



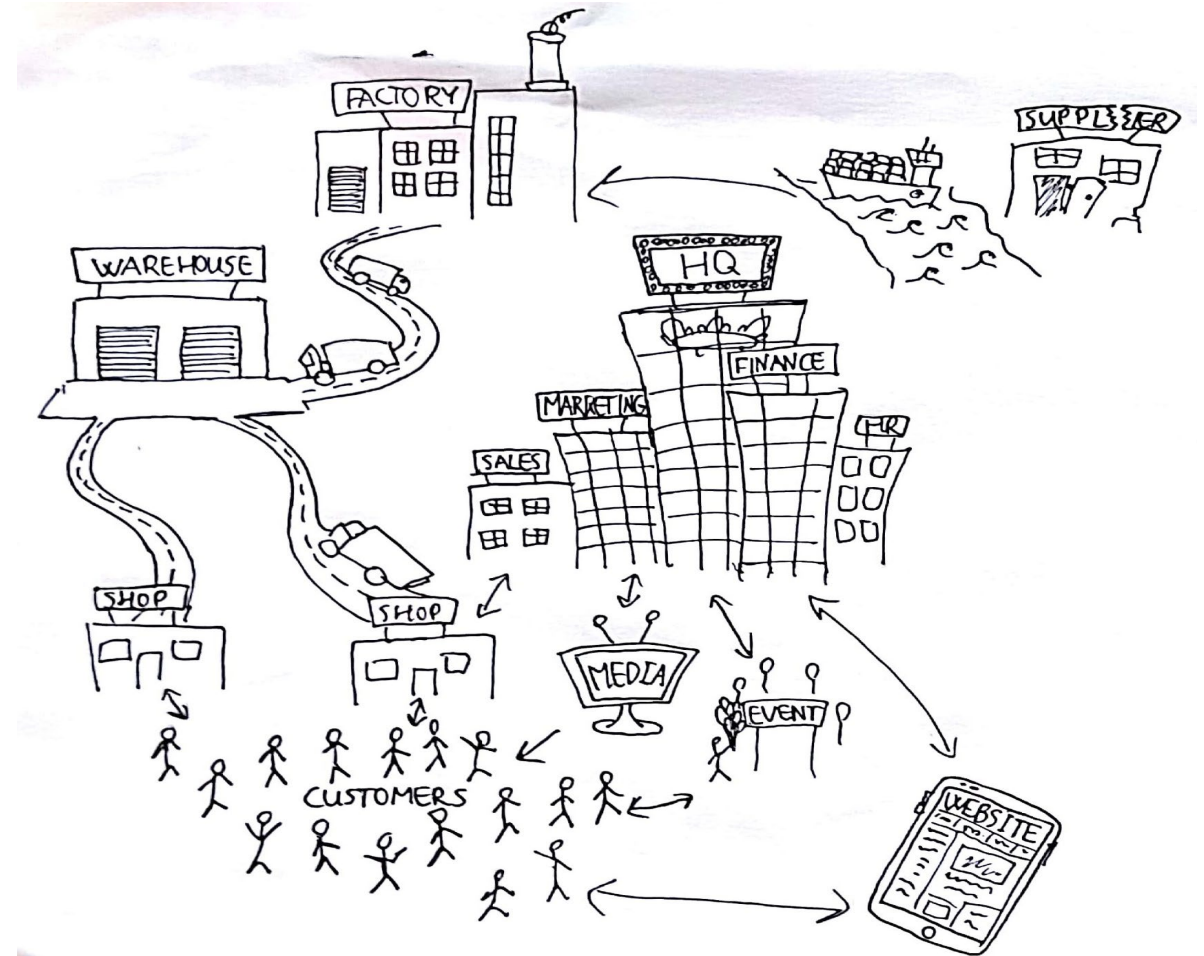
Dall-E, 2025

Examples of data centric applications

- Find out which website design make most visitors subscribe to a service (A/B testing)
- Predicting when a factory machine will break down or when a product will be returned (Predictive maintenances)
- Predicting when a customer will cancel a subscription or when an employee will quit (Churn analysis, People Analytics)
- Segmenting customers into groups to target individually (Clustering)
- Predicting if an image of a skin lesion shows sign of skin cancer (Image recognition)
- Learning the seasonality in a company's sales data (time series analysis)
- Recommending new products to buy, new series to watch, or posts to engage with (Recommender systems)

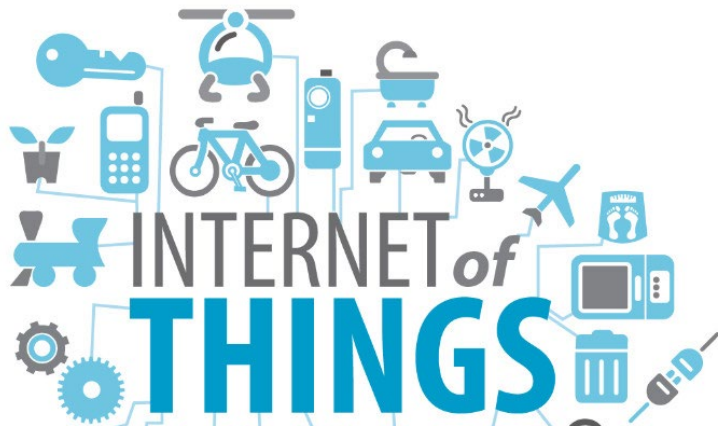
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- Every aspect of a business can be datafied....
- It helps optimize business processes, increase business understanding, develop better products, and reaching the right costumers



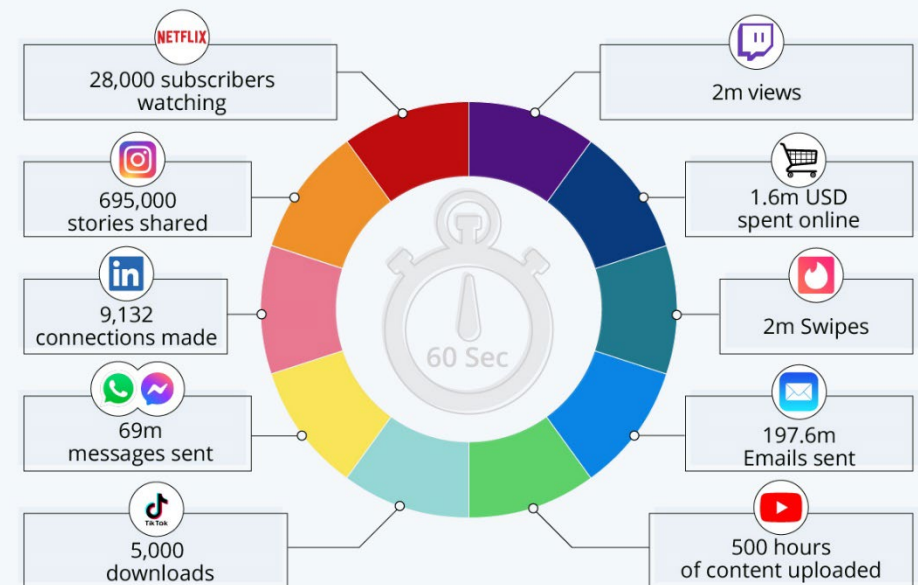
Data & Things – spring 25

- Why now?
- Where does data come from?
 - The internet (and social media)
 - Smartphones
 - Internet of Things
 - Scientific instruments



A Minute on the Internet in 2021

Estimated amount of data created on the internet in one minute



Source: Lori Lewis via AllAccess



statista

Data & Things – spring 25

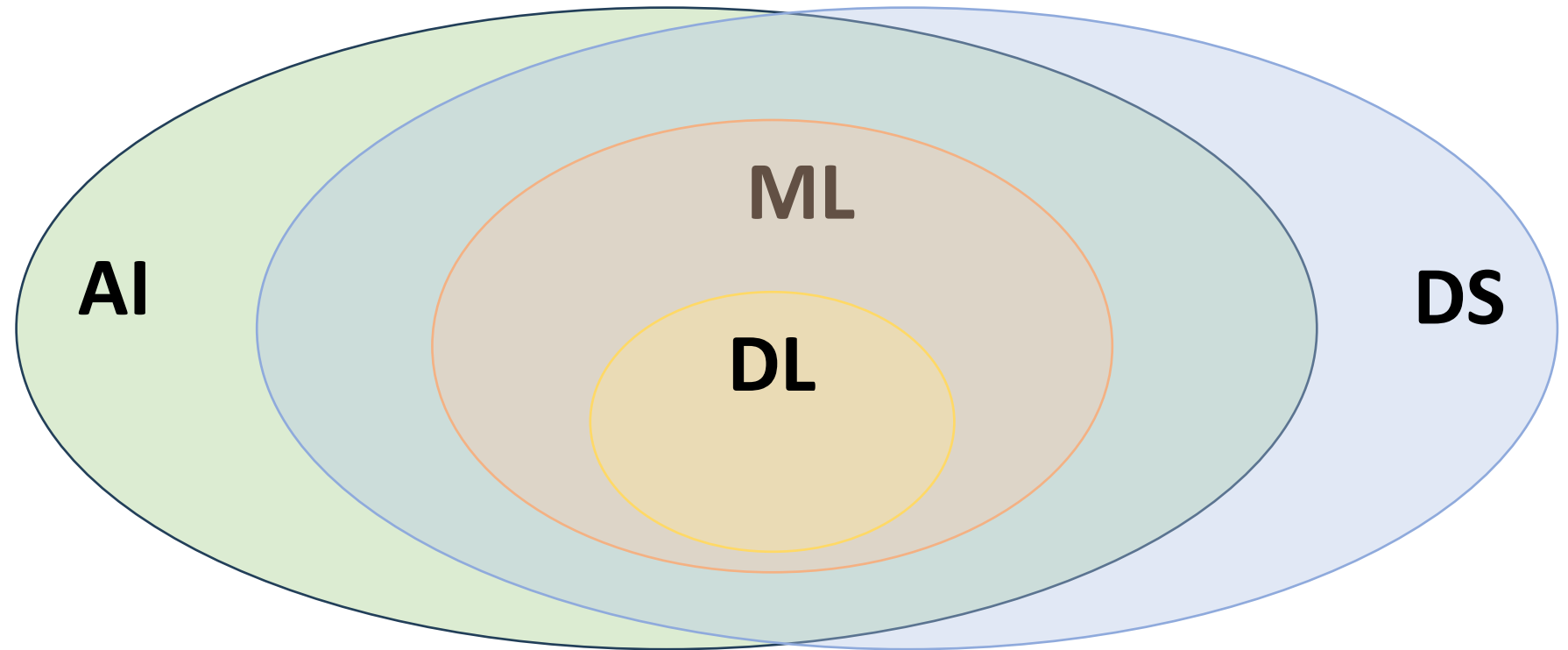
- **Getting under the hood**

- How do we collect, store and manage the necessary data?
- How do transform data to be useful?
- How do we analyze data to create insights?
- How do we build predictive models?
- How do we build and manage data centric application?
- What are the ethical and societal consequence of the data centric applications?

- All of these questions are the focus of this course, with a strong focus on analyzing data to create insights and building predictive (machine learning)models

Data & Things – spring 25

- AI: Artificial Intelligence
- DS: Data Science
- ML: Machine Learning
- DL: Deep Learning



Outline of this lecture

- Introduction to the course
- Course practicalities and the exam
- Introduction to Data Science (and Data Engineering)
- Brief introduction to Python (and Jupyter Notebook)
- Essential data science packages in Python for data science: (NumPy) and Pandas
- Examples on real data
- Exercises

Course practicalities

- **We expect you to attend all classes and show up on time!**
 - The course is planned under the expectation that you show up for all classes
 - It is your own responsibility if you choose not to show up for the classes!
- **Note the complex and condensed schedule – see study.ruc.dk**
 - Teaching will be on Monday, Wednesday, and Friday
 - Note the times differ for the different days and the different weeks!
 - The room is always here (44.3-40) except next time, where it is 46.3-042 (add the calendar from study.ruc.dk to your own calendar to keep up to date)
- **Expected workload**
 - A full-time study and no other study activities simultaneously with Data & Things
 - Total hours (from study.ruc.dk): 270 hours
 - Classes (18 * 4 hours + Q&A session): 76 hours
 - Preparation for classes (approx. 7 hours pr class): 140 hours
 - Exam and preparation before the course: 54 hours

Course practicalities

- **Course website**

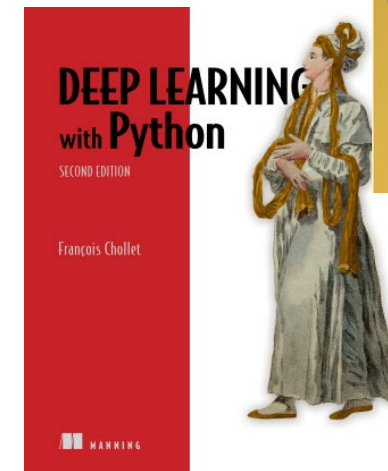
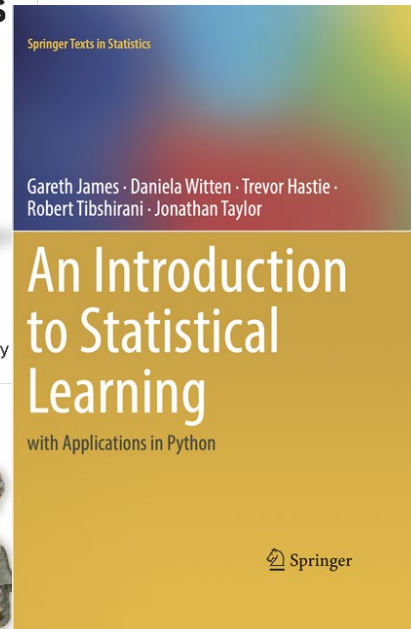
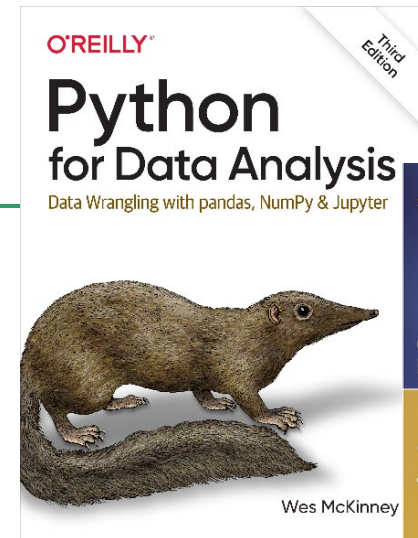
- Course material etc. all on the course moodle page:
<https://moodle.ruc.dk/course/view.php?id=23507>
- The overall topics are there already
- Readings for the classes will be posted continuously (most of it is there already)
 - Some of the later topics will be updated as we go along – so keep an eye out for updates!
- Updated room information on:
<https://study.ruc.dk/class/view/34785>

The screenshot shows the RUC (Roskilde University) website interface. At the top, there's a dark green header with the RUC logo, a 'Betjeningspanel' (Service panel), a search icon, a user profile icon, and a 'Redigeringsstilstand' (Editing status) toggle. Below the header, the main content area is titled 'Data & Things F2025' with a 'Computer Science' tag. A navigation bar includes links for 'Rum', 'Indstillinger', 'Deltagere', 'Karakterer', 'Rapporter', and 'Mere'. The main content is divided into two columns. The left column features a section titled 'Data & Things Spring 25' with a 'Sammenfold alle' (Collapse all) button. Under this, there's a 'Brief course description' and a 'Course structure' section. The right column contains a 'Nyt fra din underviser eller din uddannelse' (News from your teacher or your education) section, which includes a 'Welcome to the course' message and a 'BlueTech hackathon' announcement. The bottom of the page shows a small question mark icon and a timestamp '2025-01-17 13:58'.

Course practicalities

- **Course material**

- Different books, papers and websites...
 - We have tried to find materials that are freely available online or available online through the RUC library.
- Books we will use:
 - *Python for Data Analysis*, 3rd edition (2022) by Wes McKinney (O'Reilly): <https://wesmckinney.com/book/>
 - *An Introduction to Statistical Learning – with Applications in Python* by James, G., Witten, D., Hastie, T., Tibshirani, R., and Taylor, J., (2023), Springer (freely downloadable from: <https://www.statlearning.com/>)
 - *Data science from scratch: first principles with python*, 2nd edition (2019), O'Reilly Media by Joel Grus: <https://www.oreilly.com/library/view/data-science-from/9781492041122/>
 - Previously used book – we will only use very few chapters this year,
 - *Deep Learning with Python*, 2nd edition (2021), Manning Publishing by François Chollet: <https://www.manning.com/books/deep-learning-with-python-second-edition>
 - We will only use the first 4 chapters, approx. 120 pages. Used by the deep learning course as well
 - ...



Course practicalities

- **Prerequisite**

- We do not assume any knowledge of Python or data science, however...
 - We assume basic math skills (corresponding to level B of high school)
 - We assume that you, as a graduate student in computer science, can install Python and Jupyter Notebooks on your own and familiarize yourself with the very basics of Python – we will teach you how to use Python to do data science
 - Clearly if you have already taken Hua's elective course on Data Science and Visualization on the bachelor level, have done semester projects about data science, or taken general NatBach courses in Python and mathematics, you are likely to find the course easier – that is how it is with the RUC structure!
 - We do not assume any of these prerequisite!

Course practicalities

- **The lecturers**

- Jens Ulrik Hansen
 - Associate Professor of Computer Science, RUC
 - jensuh@ruc.dk
- Shabab Akhter
 - Manager – Data & AI, ROCKWOOL Group
 - shabab@ruc.dk



Course practicalities

• The classes

Lect.	Content	Lect.	Date
1	Intro. to course, data science, and python	Jens	Feb 3
2	Data transformation and exploratory data analysis	Jens	Feb 5
3	Data Engineering	Shabab	Feb 7
4	Statistics	Jens	Feb 10
5	Linear Regression	Jens	Feb 12
6	Time series	Shabab	Feb 12
7	Classification I	Jens	Feb 14
8	Classification II	Jens	Feb 17
9	IoT and sensor data	Shabab	Feb 19
10	Clustering	Jens	Feb 21

Lect.	Content	Lect.	Date
11	Improving and selecting ML models	Jens	Feb 24
12	Machine Learning Operations (MLOps)	Shabab	Feb 26
13	Recommender systems	Jens	Feb 26
14	Neural networks and deep learning I	Jens	Feb 28
15	Neural networks and deep learning II	Jens	Mar 3
16	Generative AI	Shabab	Mar 5
17	Explainability	Shabab	Mar 7
18	Ethical reflections on data science, end of course	Jens	Mar 10
	Exan Q&A	Jens & Shabab	Mar 12

Course practicalities

- **My classes**

- I will assume you have looked at the mandatory reading – it is up to you how careful
- I will briefly review the basic parts and put it in larger contexts, potentially going beyond the reading
- I will show examples on (real) data
- You will work with (real) data in exercises for the last 1-2 hours

Course practicalities

- **The exam**

- Exam Dates:
 - **Hand-in of written product: Sunday March 16 at 10.00am on eksamen.ruc.dk**
 - **Oral exam: Thursday March 20 and Friday March 21**
- **Format: 20 min individual oral exam based on a written product**

Course practicalities

- **The oral exam**

- Dates: March 20&21
- There will 20 minutes for each student, including time for assessment and feedback
- At the beginning of the exam, the student draws a random number. Each number will correspond to an exam topic (14 topics – see exam document on moodl).
- The student then present on the topic (3-5 min) including potential selected exercises handed in as part of the written product.
- This is followed by questions about the presentation and the exam question from the examiners (5-10 min).
- At the end, the examiners might relate their questions to some of other exam topics.
- For each of the exam topics, the student is expected to know the central concepts, methods, theories, and problems discussed in class and be able to explain and exemplify them. Moreover, for those exam topics where there are selected exercises handed in as part of the written product, the student is expected to be able to explain how they solved the exercises and be able to explain their entire code.

Course practicalities

- **The written product**

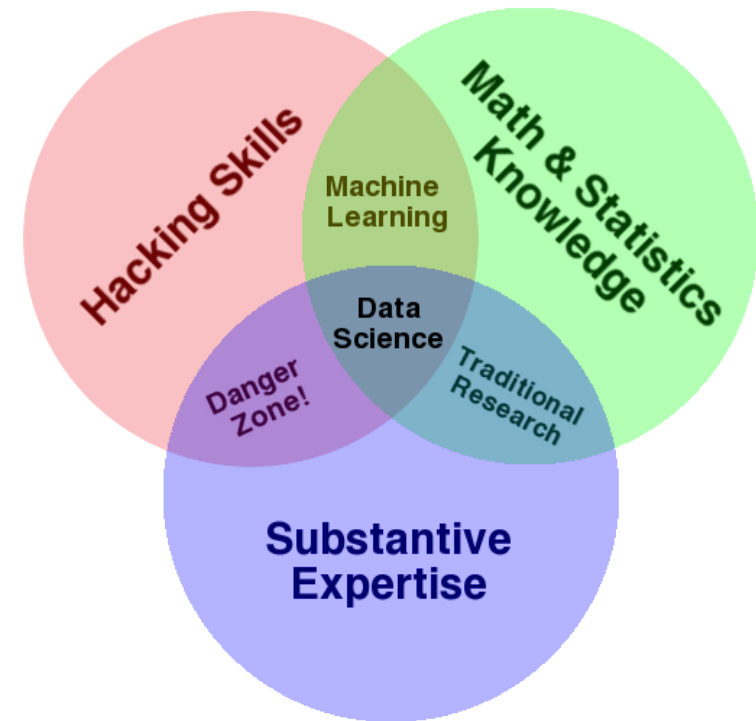
- Hand-in date: Sunday, March 16 at 10.00am
- The written product will consist of answers to selected exercises - these will be selected from those that have already been done in class. Only some of the exam topics will have such selected exercises (most of them will!).
- The list of selected exercises will be made public on the last day of class (Monday March 10).
- The handed in answers to the selected exercises should be in the format of a single Jupyter Notebook or a zip file containing a notebook for each of the selected exercises.
- The students can do the selected exercises in self-made groups and either hand in as a group on Eksamen.ruc.dk, or hand in the notebook(s) individually.

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Introduction to Data Science

- Wikipedia
 - *“Data science is an interdisciplinary academic field [1] that uses statistics, scientific computing, scientific methods, processes, algorithms and systems to extract or extrapolate knowledge and insights from noisy, structured and unstructured data.[2] Data science also integrates domain knowledge from the underlying application domain (e.g., natural sciences, information technology, medicine).[3]”,* https://en.wikipedia.org/wiki/Data_science, retrieved 2023-01-31
 - “The data science Venn diagram”, <http://drewconway.com/zia/2013/3/26/the-data-science-venn-diagram>, retrieved 2023-01-31
 - “A data scientist is someone who knows more statistics than a usual programmer and someone who knows more programming than a usual statistician”
- Me: Using data and scientific methods to solve problems and create product



Introduction to Data Science



Dall-E, 2025

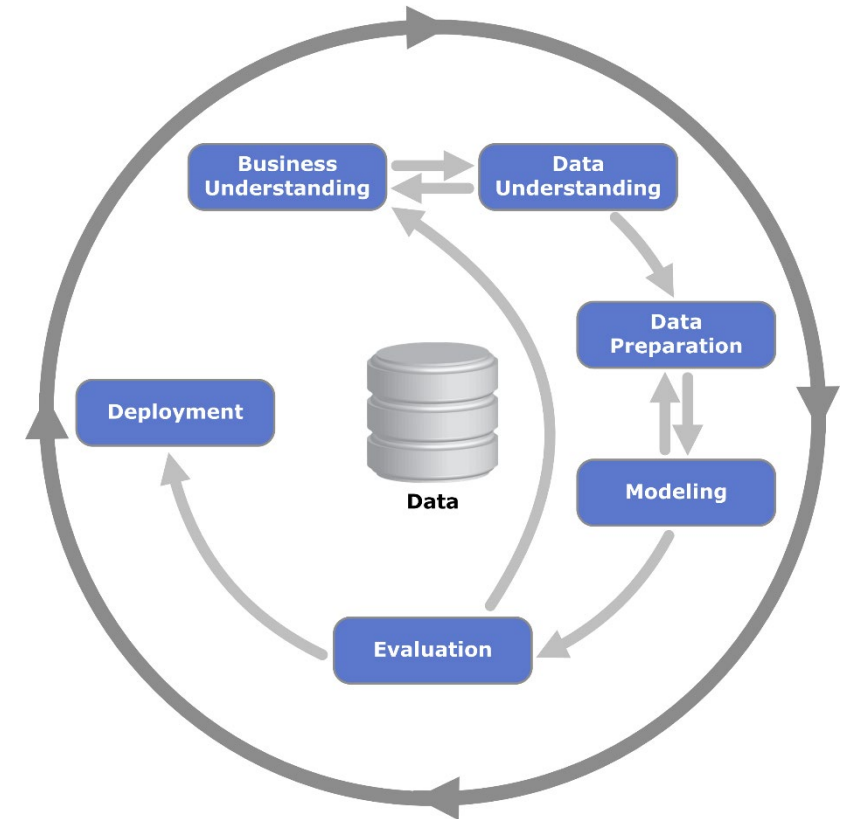
Recall the examples...

- Find out which website design make most visitors subscribe to a service (A/B testing)
- Predicting when a factory machine will break down or when a product will be returned (Predictive maintenances)
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Introduction to Data Science

- **The Data Science process**

- CRISP-DM: Cross-industry standard process for data mining
 - https://en.wikipedia.org/wiki/Cross-industry_standard_process_for_data_mining
 - From the late 90'ties, but still used today
- Data Science folklore: Data preparation/Data transformation/Data Wrangling/Data tidying take up 70% of the time in a data science project



Introduction to Data Science

- The pre-requisite for Data Science

- Data
- Data infrastructure
- ... in other words ***“Data Engineering”***
- *Data literacy* – understanding of relevant data and the problem

THE DATA SCIENCE HIERARCHY OF NEEDS

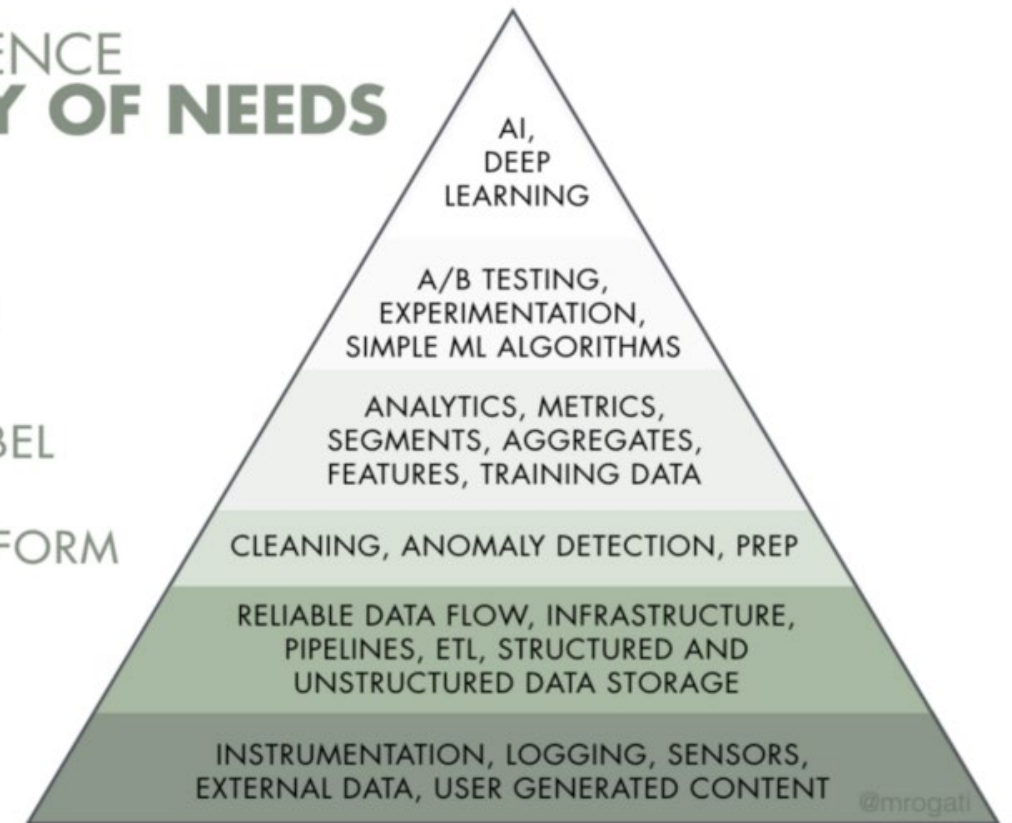
LEARN/OPTIMIZE

AGGREGATE/LABEL

EXPLORE/TRANSFORM

MOVE/STORE

COLLECT



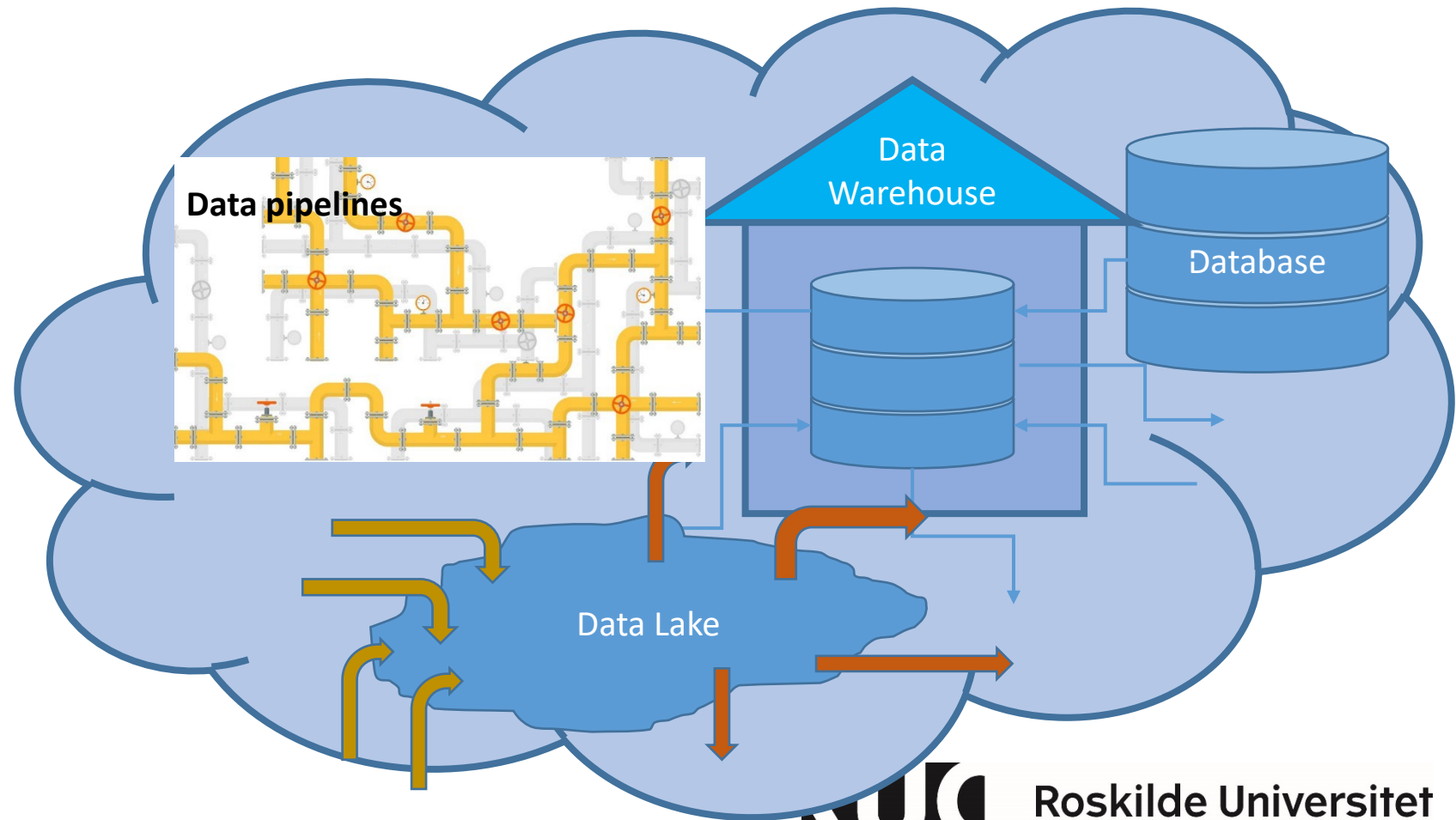
Source: Monica Rogati's fantastic Medium post [“The AI Hierarchy of Needs”](#)

Introduction to Data Science

- **The metaphors of data infrastructure**

- Database
- Data Warehouse
- Data Lake
- Data pipelines
- The cloud/laaS

- ***Data engineering:
developing and
maintaining this
data infrastructure***



Introduction to Data Science

- **The output of a data science projects:**

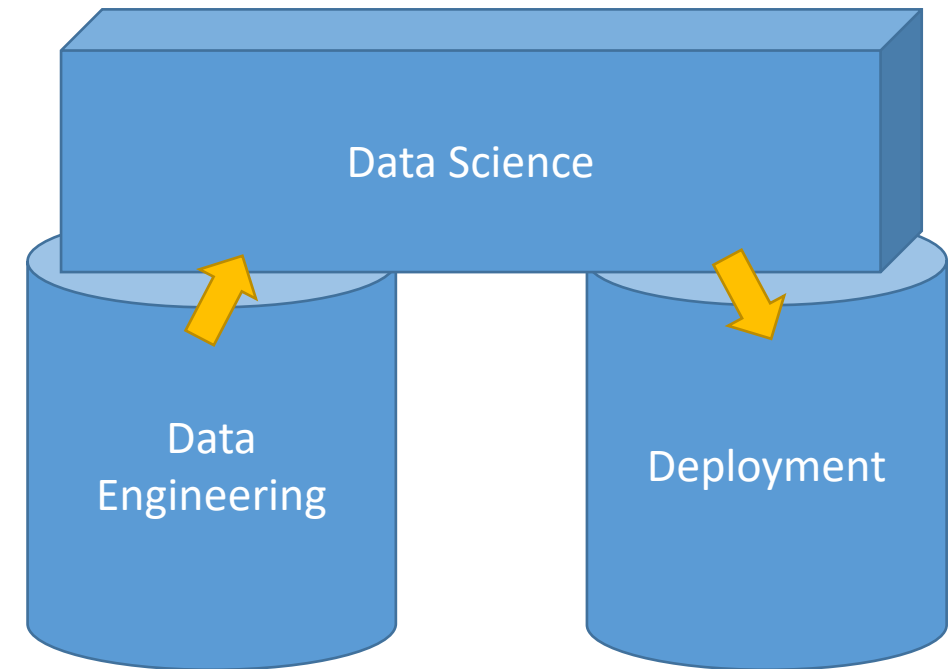
- A Visualization
- New data
- A report
- Slides
- A dashboard or interactive web-application
- An API for a predictive model
- An application involving AI/ML elements

- **Type of delivery**

- A one-off production
- A scheduled job
- A live service/application

Introduction to Data Science

- Data engineering, Data Science, and Deployment
 - The boundaries are not hard and fixed
 - Data pre-processing can be done both by data scientists or data engineers
 - Data scientists can be involved in creating data science products
 - The deployment part is often referred to as “MLObs” and are gaining more and more focus



Introduction to Data Science

- **How doing data science differs from software development**
 - Similarities
 - They both deal with data and models of the world
 - They both use programming languages
 - If a data science product (like a predictive machine learning model) is embedded in a software application, a lot of software development and CI/CD is needed.

Introduction to Data Science

- **Software development**

- In software development applications, a data model is specified as a model of the world where all relationships are specified in advance. The application then uses this to store, update and retrieve information (about customers for instance).
- (Normalized) databases are ideal for this type of dealing with data
- The data aspect of the application can be **tested** on made up test data

- **Data science**

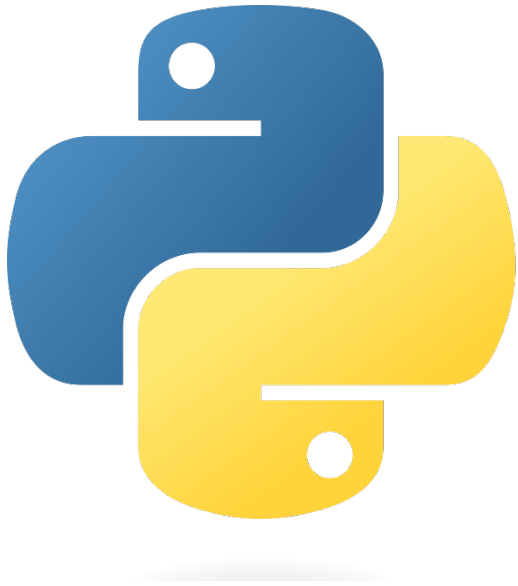
- We have no model of the world in advance, and instead of specifying one, we want to learn it from large amount of historical data about the world.
- We are not doing a lot of read, write, update, so normalized databases are not necessarily an advantage – we might as well have data in a CSV file, as we might just do one process of it to learn.
- We cannot test how good a machine learning model is on made up test data – we need real test data from the world. Moreover, we need mathematical and statistical theory about machine learning models to be able to decide how good a model we have made.

There are important differences in how we model the world, deal with data, and test our models.

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Brief introduction to Python



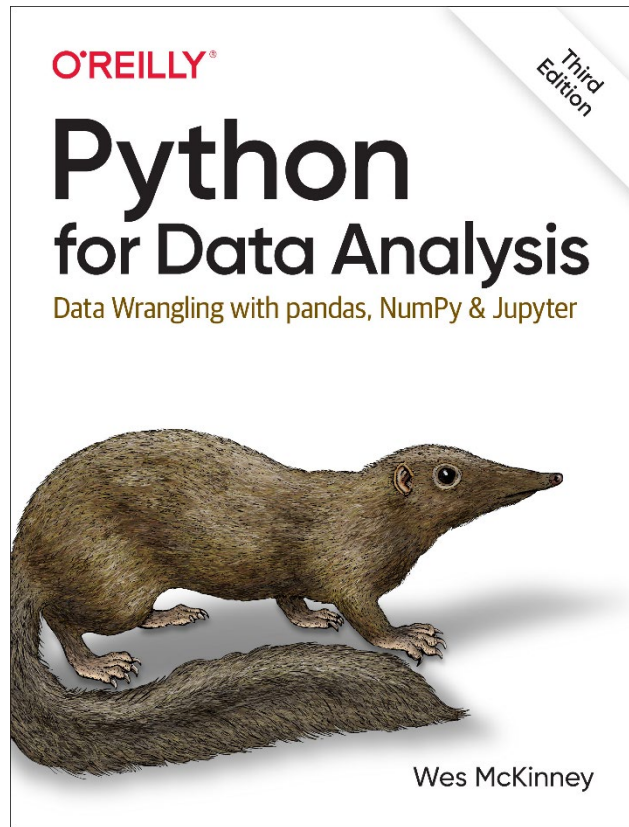
- We will use Python together with Jupyter Notebook
- We recommend installing Python through the miniconda distribution (<https://docs.conda.io/en/latest/>)
 - See Section 1.4 of Python for Data Analysis: https://wesmckinney.com/book/preliminaries#installation_and_setup
 - It will give you Python 3.11 (I think) and Jupyter Notebook
 - Alternatively, you can install the full Anaconda distribution of Python – it will give you more tools and packages and an Anaconda application
 - Alternative, you can use python notebooks in Google Colab or Visual Studio Code
- Python
 - Fairly old (early 1990ties), but very popular programming language
 - One of the two major programming languages for Data Science – the other one being R
 - General-purpose programming language with both object-oriented and functional programming.

Brief introduction to Python



- Jupyter Notebook
 - Named after Julia, Python and R.
 - Formerly IPython Notebook
 - Web-based interactive environment for creating and running notebook documents
 - A notebook is a type of document that consists of code and text cells
 - Great for learning as you can read and try out code at the same time
 - Great for communicating and sharing data science project as it works like a report that also contains all the code
 - Jupyter Notebooks can be shared and run without any problem
 - (JupyterLab is a newer and more extensive application)

Brief introduction to Python



- Starting up Jupyter notebook with Python
 - https://wesmckinney.com/book/preliminaries#installation_and_setup
 - Start the miniconda prompt (*"Anaconda Prompt (miniconda3)"*)
 - Create an environment (*"conda create -y -n pydata-book python=3.10"*)
 - Activate the environment (*"conda activate pydata-book"*)
 - Potentially install new packages (*"conda install -y pandas jupyter matplotlib scikit-learn"*)
 - Starting Jupyter Notebook (*"jupyter notebook"*)
 - Go through example notebooks (from moodle) if you are not familiar with Python and Jupyter notebooks
 - A Crash Course in Jupyter Notebook.ipynb
 - A Crash Course in Python.ipynb

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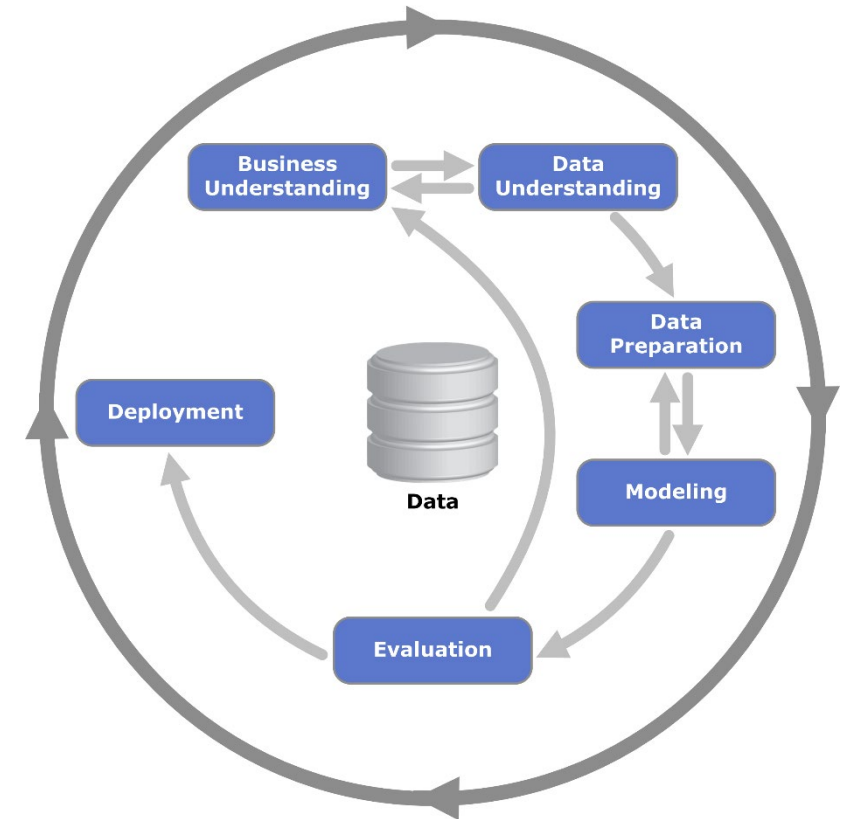
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Essential data science packages in Python for data science: (NumPy) and Pandas

- Some of the most essential data science packages in Python
 - **NumPy** (<https://numpy.org/>)
 - A library for optimized numerical computations and advanced tensor data structures
 - **pandas** (<https://pandas.pydata.org/>)
 - Introduces the DataFrame data structure along with tools for reading data into DataFrames, as well as tools for manipulating, transforming and summarizing DataFrames.
 - **Matplotlib** (<https://matplotlib.org/>)
 - A classic library for making data visualizations in Python
 - **Seaborn** (<https://seaborn.pydata.org/>)
 - Another plotting library that builds on top of matplotlib to provide nicer visualizations and integration with pandas.
 - **Scipy** (<https://scipy.org/>)
 - Extent NumPy with more functionality to make scientific computations including basic statistics
 - **Statsmodels** (<https://www.statsmodels.org/stable/index.html>)
 - A library for doing statistical modeling in Python
 - **scikit-learn** (<https://scikit-learn.org/stable/>)
 - The classic library in Python to do machine learning
 - ...

Essential data science packages in Python for data science: (NumPy) and Pandas

- Back to the data science process...
 - *“Data Science folklore: Data preparation/Data transformation/Data Wrangling/Data tidying take up 70% of the time in a data science project”*
 - Being skilled in this part can really make a difference in the success and speed of data science projects!
 - It is hard to teach! (at least without being somewhat boring)
 - We will talk a bit about it today and next time, but you will only properly learn these skills by getting your hands dirty by working on real data (- thus, you will improve your skills throughout the course)



Essential data science packages in Python for data science: (NumPy) and Pandas

- NumPy
 - Has a basic data structure called ndarray that are highly useful multi dimensional arrays (also useful to represent the tensors used in Deep Learning)
 - Allow for efficient multiple computation without the need of for loops
 - See the notebook “NumPy introduction.ipynb”

Essential data science packages in Python for data science: (NumPy) and Pandas

- Pandas
 - Works with tabular heterogeneous data (contrary to NumPy that works with homogeneously data of type numerical).
 - Has basic data type pandas Series and pandas DataFrame
 - DataFrame is a central data structure in doing data science on structured data
 - Pandas also provide functions for reading data into DataFrames
 - See the notebook “pandas basic.ipynb”

Essential data science packages in Python for data science: (NumPy) and Pandas

- **Good steps for loading data into Python from a file**

- 1) Figure out what format the file is
- 2) Figure out the right data reading function to load that format
- 3) Have a look at the file (in Excel, LibreOffice, a text editor, ... etc.)
- 4) Determine whether there are any specific arguments you need to provide to the data reading function (separator, skipping rows, ... etc.)
- 5) When you have loaded the data into Python, have a look at it (using head, tail, shape, etc.)
- 6) Is the data as expected? If not, repeat step 3-6.

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Examples on real data

- Let us look at some real examples
 - The notebook “first data examples.ipynb”

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Exercises

- Make sure you have Python and Jupyter notebook installed!
- If you are unfamiliar with Python or Jupyter notebook, you can look at the notebooks “A Crash Course in Jupyter Notebook.ipynb” and “A Crash Course in Python.ipynb” on moodle.
- **Do the exercise at the end of the “first data examples.ipynb” notebook**
- (Load the IRIS dataset from scikit-learn – what is the data all about?)
- (Find an interesting dataset on UCI Machine Learning Repository and load it into a Python notebook and investigate it)